

A1 on a 3+1D Diamond Waist: the Area Coefficient is UV-Stable; the Unsubtracted Sea is Not Mean-Zero Curvature

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Abstract

The same A1 logic as Paper 21, now an area law on the diamond waist of Papers 22–23. Peschel entropy $S = -\text{Tr}[C_A \log C_A + (1 - C_A) \log(1 - C_A)]$, lattice area $A_{\text{cut}} = \text{bonds leaving the ball}$, $S = \alpha A_{\text{cut}} + \beta$.

$N = 16$, $R = 2, 3, 4, 5$: $\alpha(0) = 0.245$. UV masses with $mR_{\text{max}} \leq 0.5$ shift α by at most 4.3%. Area-law RMSE 0.27 versus volume-law 2.96. Auditor, $N = 14$, $R = 2, 3, 4$: $\alpha(0) = 0.245$, relative drift 2.1%. C2 PASS. [P] as a necessary-condition test on this field.

The unsubtracted sea density is $\varepsilon = -0.970$ per site. That is not mean-zero curvature. It is not quantum foam. It is not $\eta = 1/4G$.

1 Gate

Locked before the run. C1: $\alpha(0) > 0$. C2: relative UV drift $< 20\%$ for $mR_{\text{max}} \leq 0.5$. C3: area fit beats volume fit at $m = 0$. C4: at least three radii.

2 Numbers

mR_{max}	α	rel to $m = 0$	window
0	0.245	0	—
0.10	0.244	0.24%	UV
0.25	0.241	1.4%	UV
0.50	0.234	4.3%	UV
2.0	0.190	23%	IR, not a gate

The coefficient moves when the mass is no longer UV. Inside the locked window it does not. That is A1 for this area law.

An independent adversary on $N = 13$ (odd, 37 exact zero modes) found $\alpha(0) = 0.245$ and a worst UV drift of 5.9%. C2 stands. Catch: C3 is observable-dependent. Cut bonds beat volume. Boundary-site count and $4\pi R^2$ need not. Absolute α depends on the area proxy. The locked proxy was cut bonds.

3 Foam and mean curvature

The destination I want is a quantum foam whose *mean* curvature vanishes. That is not a result of this note.

If the mean is an infrared statement, it is $\Lambda = 0$ and the vacuum is Minkowski. That is not de Sitter (Paper 19 needs $\Lambda > 0$). If the mean is only of ultraviolet fluctuations about a smooth background, it does not select Λ .

The free lattice sea has $\varepsilon \simeq -1$ in lattice units. Identifying that $\langle T_{00} \rangle$ with Einstein's source is the cosmological-constant problem, not a foam theorem. Setting $\langle : T : \rangle = 0$ by normal ordering inserts a zero mean by definition. I will not call that a derivation.

4 What this does not do

It does not produce $\eta = 1/4G$. It does not make Jacobson 2016 a four-dimensional theorem. It does not copy FGHMV onto de Sitter. It does not finish the foam. A2 (Paper 23) plus this A1 are necessary-condition tests on a free 3+1D fermion. They are not victory.

Admission. *I ran the A1 that the 3+1D path required. I recorded that the sea is not mean-zero curvature. I did not write that the foam is done, and I did not write that entanglement gravity is de Sitter.*